

MUROTOMISAKI, Japan

WMO: 478990

Lat: 33.2522N

Lon: 134.1772E

Elev: 613

StdP: 14.37

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.2	36.3	14.5	11.7	36.8	17.4	13.5	38.5	50.0	47.3	45.3	46.2	20.9	310	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	85.5	79.0	84.2	78.5	82.9	77.7	80.3	83.6	79.5	82.5	78.8	81.5	11.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.5	156.7	82.2	78.7	152.6	81.4	77.9	148.5	80.7	44.5	83.8	43.7	82.6	42.9	81.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.1	38.6	32.7	DB	30.3	87.8	2.5	1.4	28.5	88.8	27.1	89.6	25.7	90.3	23.9	91.3
			WB	24.8	81.8	2.2	1.0	23.2	82.6	21.9	83.2	20.7	83.8	19.1	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	45.7	46.9	51.8	59.4	66.3	71.0	77.7	79.9	75.6	68.0	59.2	50.3	(6)
		DBStd	12.43	4.88	5.51	5.38	4.59	3.13	3.22	2.93	2.34	3.48	4.11	5.26	5.55	(7)
		HDD50	381	149	118	44	2	0	0	0	0	0	2	66		(8)
		HDD65	2370	597	508	408	174	23	1	0	0	18	184	457		(9)
		CDD50	5027	16	30	100	283	506	629	858	928	768	557	277	75	(10)
		CDD65	1541	0	0	0	4	64	180	393	463	318	110	9	0	(11)
		CDH74	8866	0	0	0	0	16	287	2685	4095	1663	118	2	0	(12)
		CDH80	1245	0	0	0	0	0	6	344	739	155	1	0	0	(13)
(14)	Wind	WSAvg	16.1	17.5	16.9	17.3	16.3	15.4	15.5	15.0	14.3	15.8	16.6	15.9	16.9	(14)
(15)	Precipitation	PrecAvg	102.2	3.6	4.8	6.9	7.7	10.0	13.8	11.6	10.6	14.2	9.0	6.7	3.5	(15)
(16)		PrecMax	135.3	9.1	9.8	12.3	16.9	19.3	24.3	25.2	29.1	27.7	22.7	17.7	7.3	(16)
(17)		PrecMin	63.4	0.9	1.6	2.7	3.4	2.7	5.7	4.0	2.3	2.9	2.1	1.1	0.3	(17)
(18)		PrecStd	18.9	2.0	2.3	2.4	3.3	4.4	4.8	5.0	6.9	6.6	4.4	3.9	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	60.8	63.6	66.1	71.4	75.8	80.8	86.2	88.0	85.7	79.0	72.3	65.7		(19)
(20)		0.4% MCWB	57.4	60.6	60.6	64.4	67.5	76.3	79.3	79.8	78.6	73.8	67.4	62.2		(20)
(21)		2% DB	57.9	60.5	63.8	69.1	74.0	78.3	84.6	86.2	83.5	76.8	69.7	62.4		(21)
(22)		2% MCWB	53.9	57.3	59.3	63.2	67.0	74.4	78.7	79.3	77.6	72.2	64.6	57.6		(22)
(23)		5% DB	55.2	57.8	62.0	67.4	72.6	76.9	83.4	84.9	81.7	75.2	68.1	60.0		(23)
(24)		5% MCWB	50.0	54.2	57.5	62.1	66.4	73.6	78.1	78.8	76.1	70.1	63.2	54.7		(24)
(25)		10% DB	52.9	55.1	60.1	65.8	71.1	75.6	82.0	83.7	80.2	73.6	66.5	57.9		(25)
(26)		10% MCWB	47.2	50.4	55.3	60.4	65.7	72.8	77.4	78.3	75.1	68.1	61.5	52.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	59.3	62.0	63.8	68.3	72.4	77.4	81.0	81.8	80.4	76.6	70.0	63.7		(27)
(28)		0.4% MCDB	60.4	62.7	64.4	69.1	73.3	78.9	84.1	85.3	83.2	77.6	71.1	64.9		(28)
(29)		2% WB	55.4	58.3	61.2	66.0	70.6	76.1	79.9	80.8	78.9	74.1	66.7	59.3		(29)
(30)		2% MCDB	57.2	60.0	62.6	67.6	71.7	77.4	82.8	84.2	81.6	75.7	68.3	61.1		(30)
(31)		5% WB	51.3	55.1	58.7	63.9	69.0	75.0	79.0	80.0	77.8	71.8	64.5	56.0		(31)
(32)		5% MCDB	54.0	57.4	61.1	65.9	70.7	76.1	81.8	83.2	80.3	74.0	67.1	59.0		(32)
(33)		10% WB	48.0	51.4	56.3	62.0	67.4	73.8	78.1	79.2	76.8	69.5	62.5	53.3		(33)
(34)		10% MCDB	52.2	54.4	59.3	64.7	69.8	75.1	80.7	82.2	79.1	72.5	65.8	57.3		(34)
(35)	Mean Daily Temperature Range	MDBR	8.9	9.6	10.1	9.0	8.1	6.6	6.7	7.2	7.0	7.1	7.8	8.4		(35)
(36)		5% DB MCDBR	10.2	10.4	10.5	9.6	9.2	7.7	8.1	8.4	8.1	7.8	8.0	9.1		(36)
(37)		5% DB MCWBR	11.2	11.1	10.8	8.9	7.8	5.6	5.3	5.3	5.7	7.1	8.5	9.8		(37)
(38)		5% WB MCDBR	9.8	10.4	10.1	8.5	7.8	6.4	7.6	7.8	7.8	7.3	7.6	8.9		(38)
(39)		5% WB MCWBR	12.1	12.1	11.8	9.0	7.6	5.2	5.2	5.2	5.7	7.4	9.1	11.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.379	0.424	0.478	0.503	0.514	0.527	0.521	0.495	0.469	0.442	0.404	0.373		(40)
(41)		taud	2.258	2.144	2.001	1.988	2.039	2.087	2.151	2.224	2.266	2.250	2.273	2.295		(41)
(42)		Ebn at Noon	259	258	253	252	251	246	247	252	253	250	248	253		(42)
(43)		Edh at Noon	35	43	54	56	54	52	48	44	41	38	34	32		(43)
(44)	All-Sky Solar Radiation	RadAvg	836	1019	1327	1544	1622	1381	1656	1711	1341	1080	852	753		(44)
(45)		RadStd	66	83	96	169	171	173	182	141	116	116	78	50		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	+1.73	N/A	+0.72	+0.85	N/A	N/A	N/A	N/A
	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page